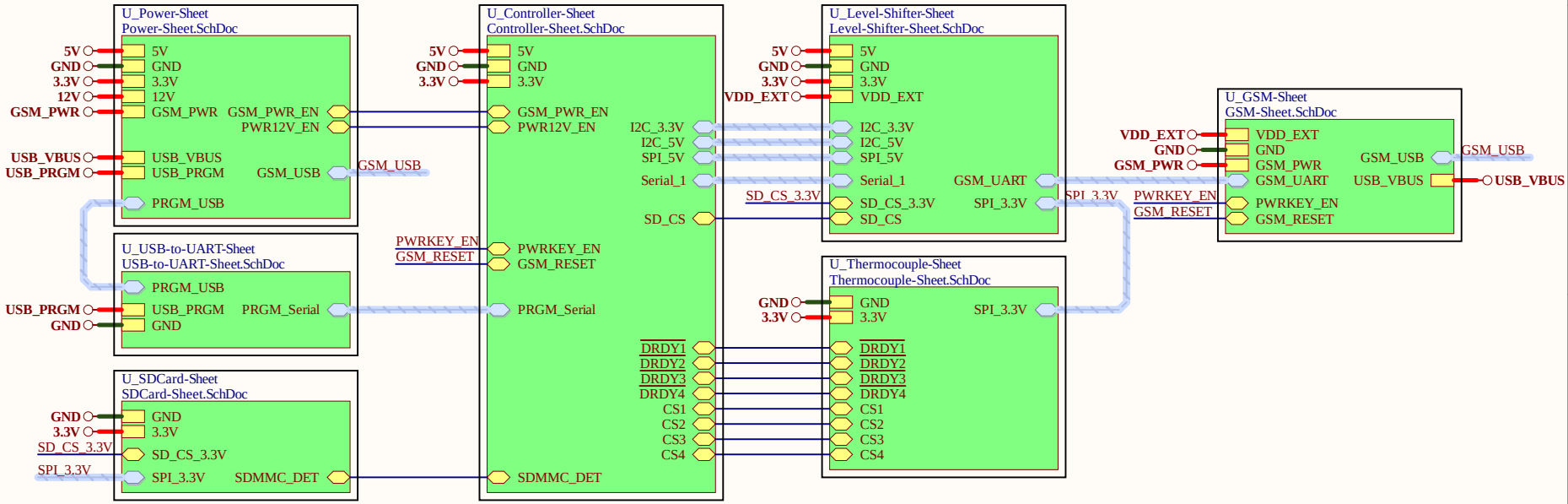
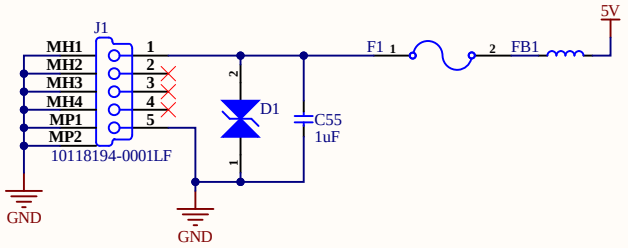


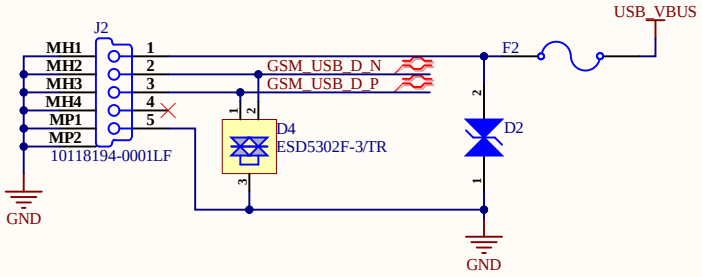
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Date: 5/18/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8

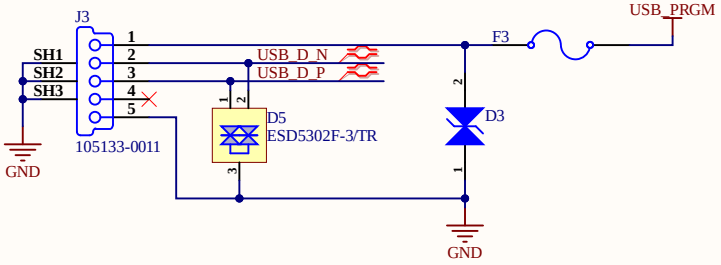
USB FOR BOARD POWER



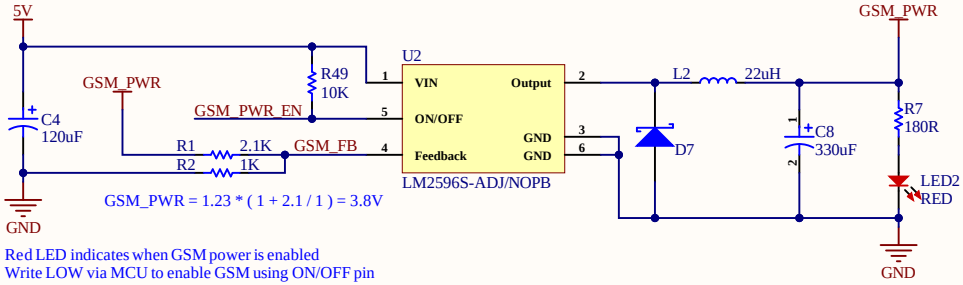
USB FOR GSM UPDATES



USB PROGRAMMING MEGA2560

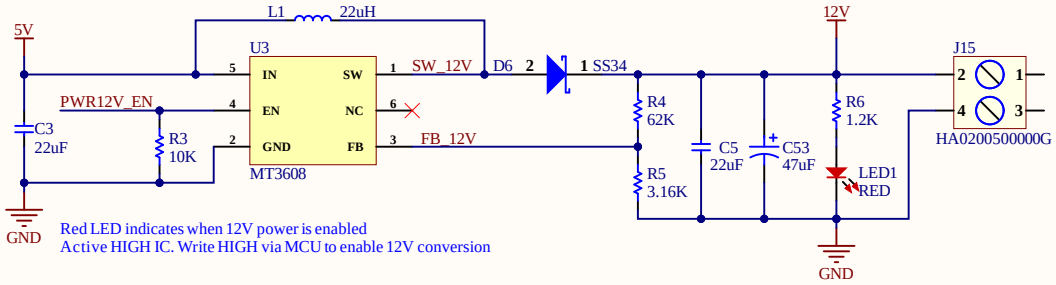


BUCK FOR GSM POWER 5V TO 3.8V



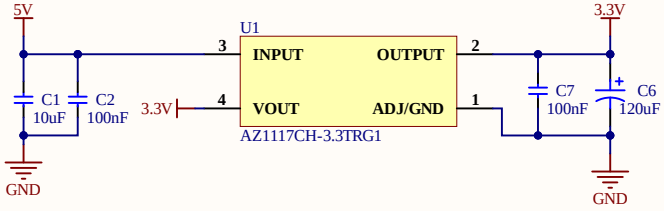
Red LED indicates when GSM power is enabled
Write LOW via MCU to enable GSM using ON/OFF pin

BOOST FOR WIKA SENSOR 5V TO 12V

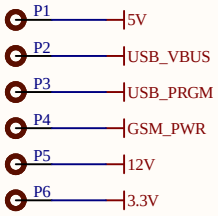


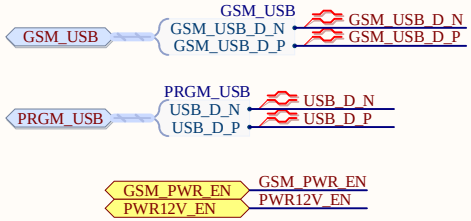
Red LED indicates when 12V power is enabled
Active HIGH IC. Write HIGH via MCU to enable 12V conversion

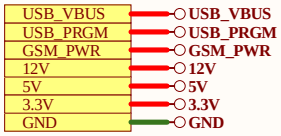
LDO 5V TO 3.3V



Test Points



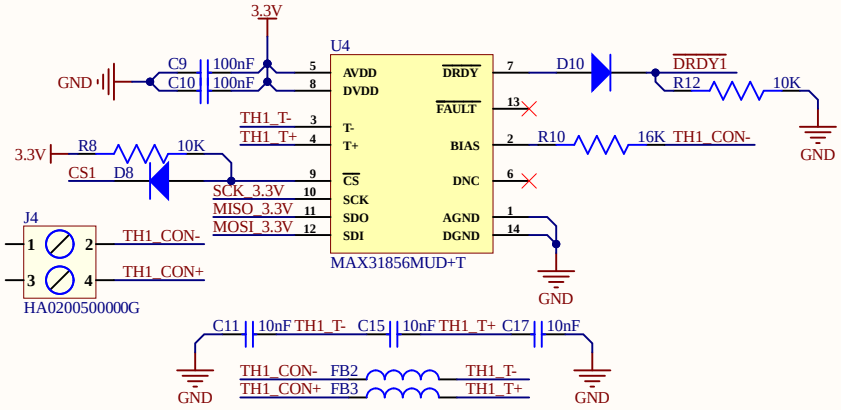




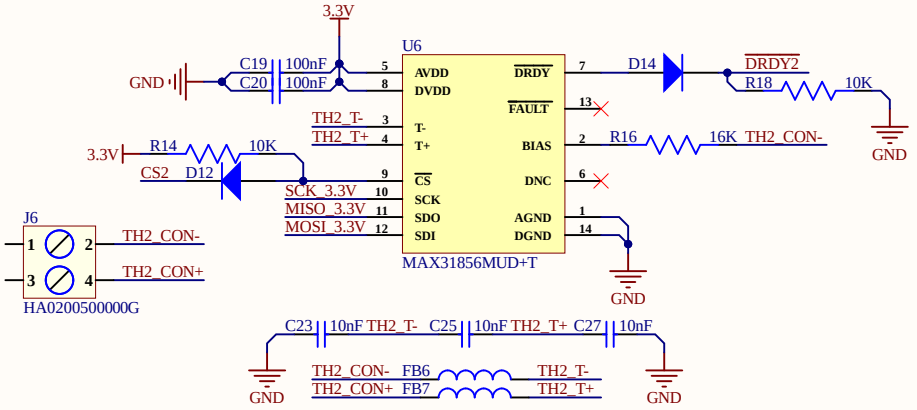
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Date: 5/18/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8

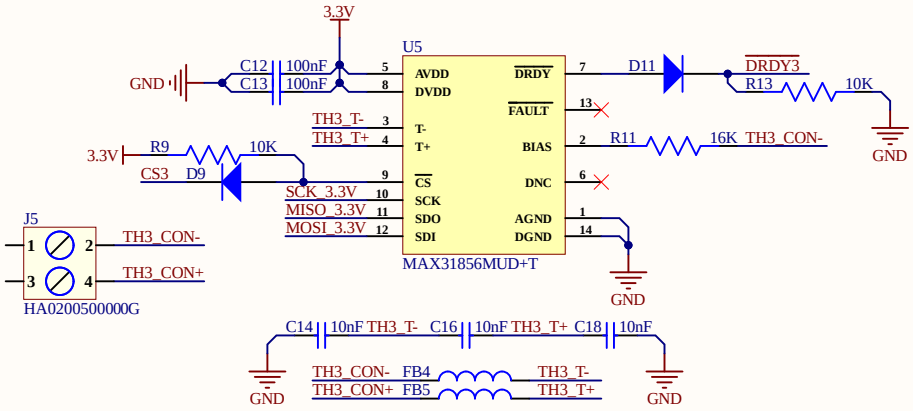
Thermocouple Input - 1



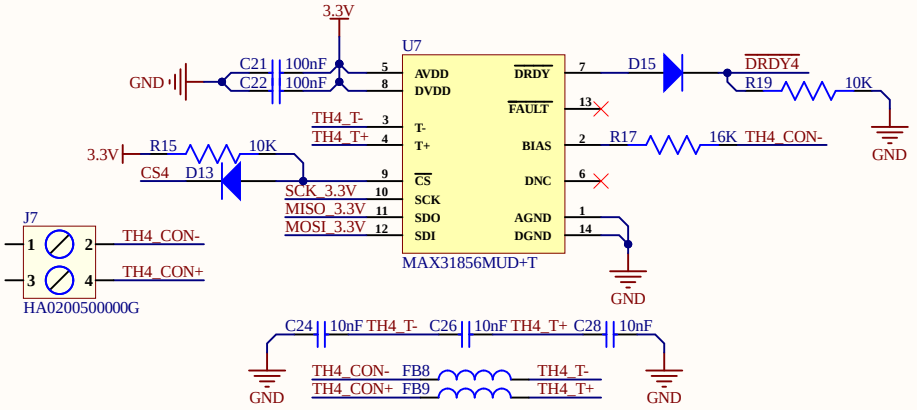
Thermocouple Input - 2



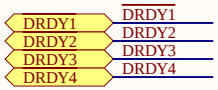
Thermocouple Input - 3

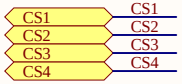


Thermocouple Input - 4







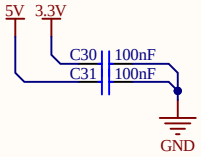


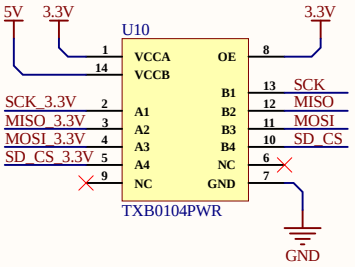


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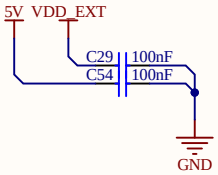
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Date: 5/18/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 8

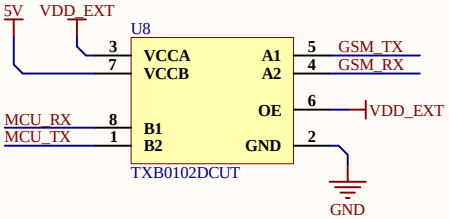
Level Shifter for SPI



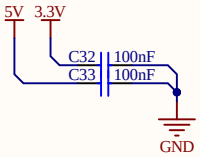


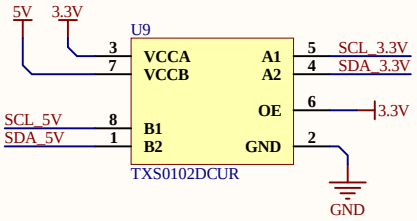
Level Shifter for Serial

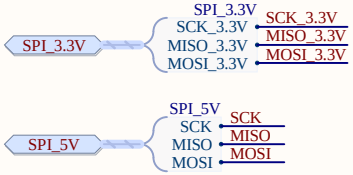


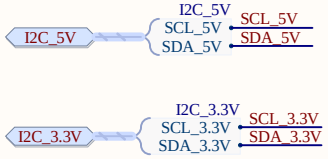


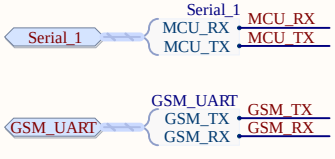
Level Shifter for I2C

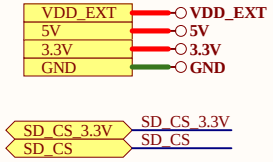








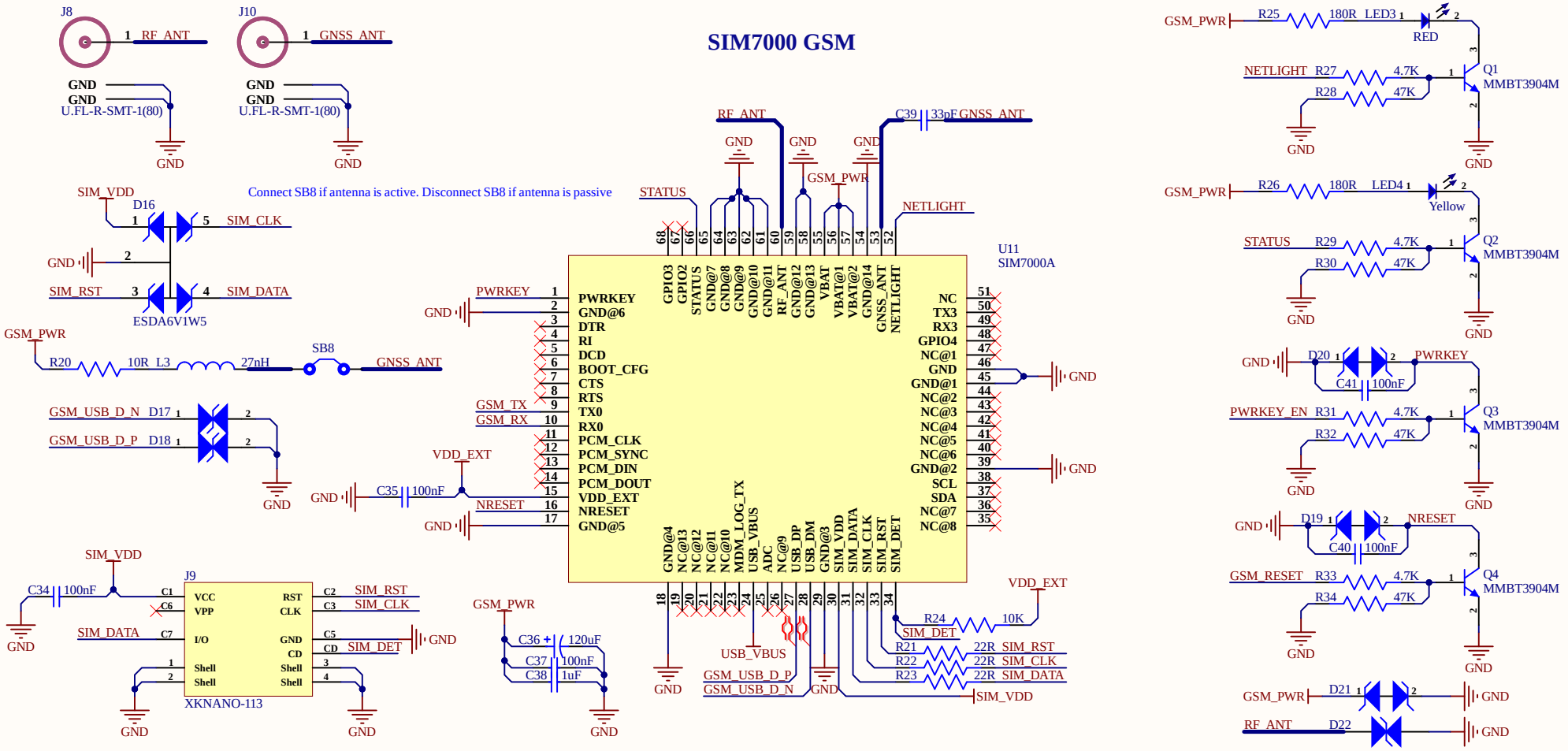




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File: Mega2560_Custom_v2.0.PrjPcb		
Date: 5/18/2023	Page	Num: 4
Drawn By: Huzaifa Ali	Page	Total: 8

SIM7000 GSM



GSM_UART
GSM_TX
GSM_RX

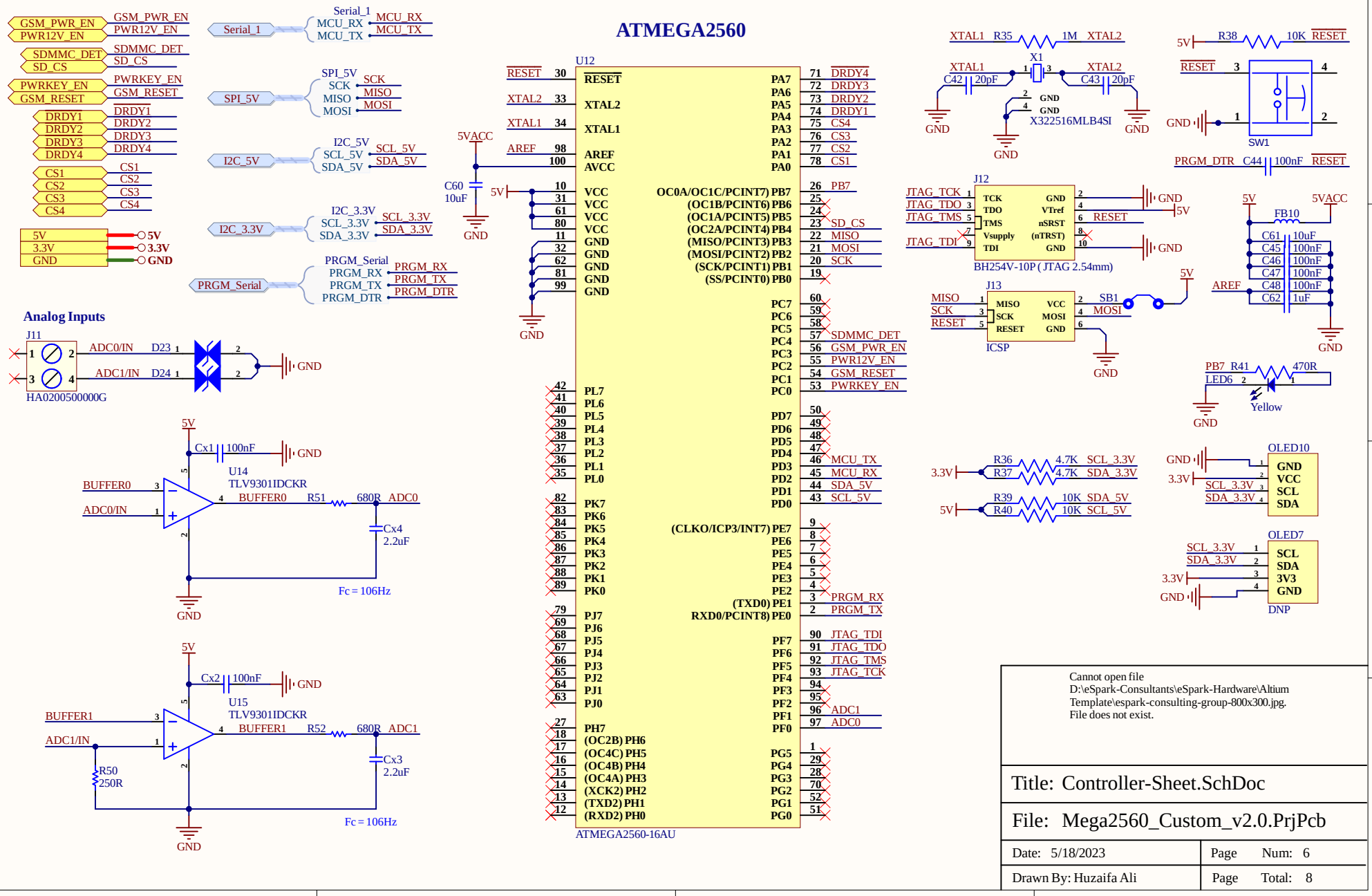
GSM_USB
GSM_USB_D_N
GSM_USB_D_P

PWRKEY_EN
GSM_RESET

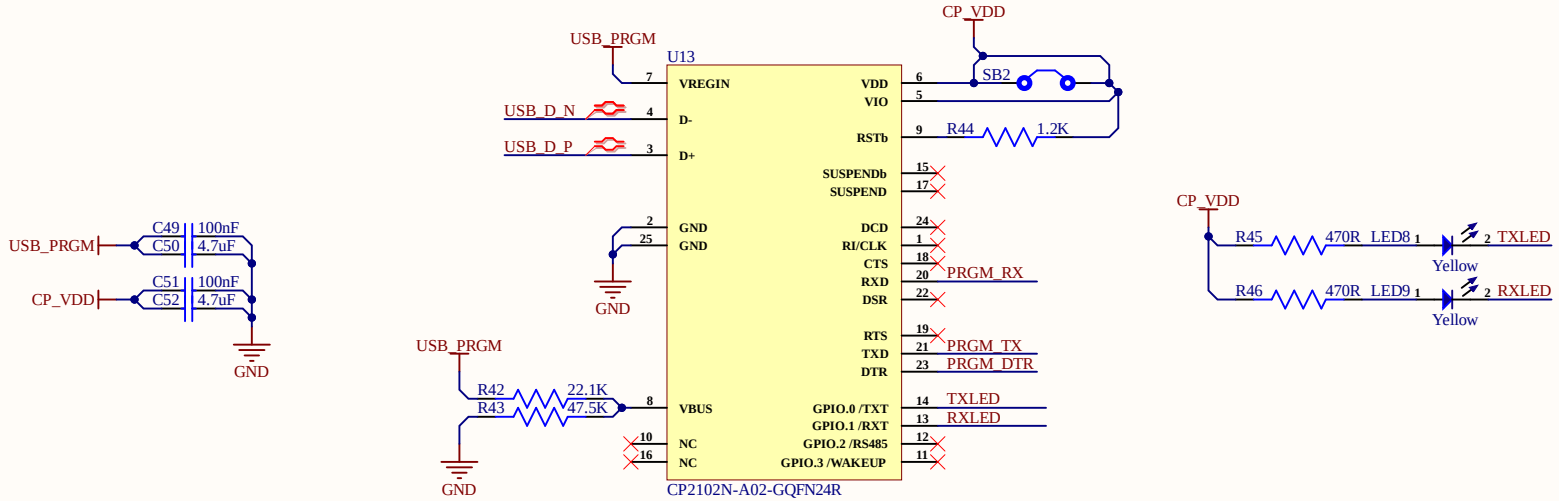
USB_VBUS
GSM_PWR
VDD_EXT
GND

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File: Mega2560_Custom_v2.0.PrjPcb	
Date: 5/18/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8


ATMEGA2560
U12
ATMEGA2560-16AU
GSM_PWR_EN
PWR12V_EN
SDMMC_DET
SD_CS
PWRKEY_EN
GSM_RESET
DRDY1
DRDY2
DRDY3
DRDY4
CS1
CS2
CS3
CS4
5V
3.3V
GND
Serial_1
MCU_RX
MCU_TX
SPI_5V
SCK
MISO
MOSI
I2C_5V
SCL_5V
SDA_5V
I2C_3.3V
SCL_3.3V
SDA_3.3V
PRGM_Serial
PRGM_RX
PRGM_TX
PRGM_DTR
Analog Inputs
J11
HA0200500000G
ADC0/IN
ADC1/IN
U14
TLV9301IDCKR
U15
BUFFER0
BUFFER1
R51
R52
680R
Cx1
Cx2
Cx4
Cx3
100nF
2.2uF
R50
250R
Fc = 106Hz
5VACC
C60
10uF
AREF
XTAL1
XTAL2
RESET
X1
X322516MLB4SI
C42
C43
20pF
SW1
PRGM_DTR
C44
J12
BH254V-10P (JTAG 2.54mm)
JTAG_TCK
JTAG_TDO
JTAG_TMS
JTAG_TDI
J13
ICSP
FB10
C61
C45
C46
C47
C48
C62
10uF
100nF
1uF
PB7_R41
470R
LED6
Yellow
R36
R37
R39
R40
4.7K
10K
SCL_3.3V
SDA_3.3V
SDA_5V
SCL_5V
OLED10
OLED7
SCL
SDA
3V3
DNP
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Date: 5/18/2023
Page Num: 6
Drawn By: Huzaifa Ali
Page Total: 8

USB-to-UART for Programming





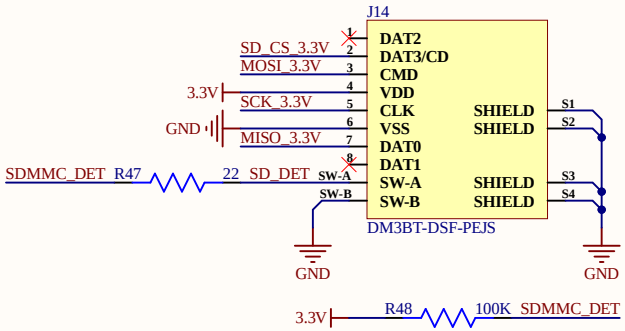


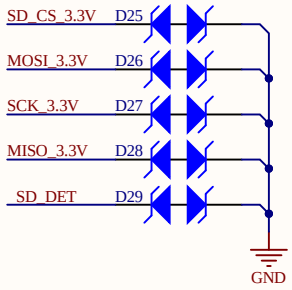


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Date: 5/18/2023	Page Num: 7
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SD Card Interface





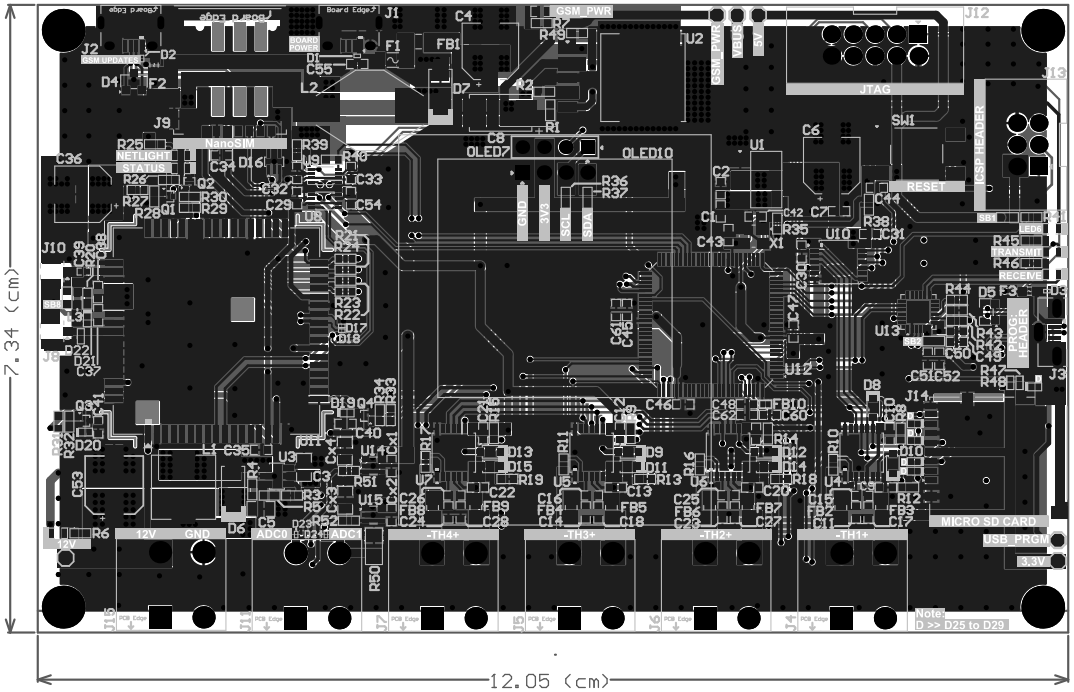






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File: Mega2560_Custom_v2.0.PrjPcb	
Date: 5/18/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 8



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	3.00mil	3.5	
1	Top Layer		10.00mil		
	Dielectric 1	FR-4	37.00mil	4.8	
2	Bottom Layer		10.00mil		
	Bottom Solder	Solder Resist	3.00mil	3.5	
	Bottom Overlay				